Lecture 23: ARMA Models

1 ARMA Model

The generalized version of the ARMA model with parameters p and q is

$$r_t = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i} + a_t - \sum_{i=1}^{q} \theta_i a_{t-i}. \tag{1}$$

For any time series x_t, introduce the characteristic functions

$$\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i, \qquad \theta(z) := 1 - \sum_{j=1}^{q} \theta_j z^j,$$

for any $z \in \mathcal{C}$. Recall that back-shift operator $Bx_t := x_{t-1}$. Then, we can rewrite the $ARMA(p,q)$ model as

$$\phi(B) r_t = \phi_0 + \theta(B) a_t,$$

or, in an extended form, as

$$\left(1 - \phi_1 B - \ldots \phi_p B^p\right) r_t = \phi_0 + \left(1 - \theta_1 B - \ldots - \theta_q B^q\right) a_t,$$

We require that $\phi(\cdot)$ and $\theta(\cdot)$ have no common factors, otherwise the order (p,q) can be reduced. In other words, we require that the polynomial $\phi(\cdot)$ and $\theta(\cdot)$ are co-prime.

Remark 1.1. *Some of the material presented in these notes is based on fundamental concepts from Complex Analysis. We refer to the online book, Taylor, Michael. Introduction to Complex Analysis, for an in-depth coverage.*

We will use the fact that a holomorphic function, i.e., one that is complex differentiable in a neighborhood of each point in its domain admits a power series representation within its domain.

Stationarity: The $ARMA(p,q)$ model defines a stationary, linear process if and only if all the roots of the AR characteristic equation $\phi(z) = 0$ are not on the unit circle in the complex plane.

Causality: The requirement that ϕ has no complex zeros inside or on the unit circle guarantees *causality*, i.e., that the return at time t depends on present and past innovation (shock) terms a_s, $s \leq t$, but not on future innovations. We also say that the $AR(p)$ process is causal.

Invertibility. We require that θ has no complex zeros inside or on the unit circle guarantees that the ARMA model is *invertible*, i.e., we can recover the innovation terms a_t using present and past observations of the returns r_u, $u \leq t$.

2 MA Representation of ARMA(p,q) models

Given two polynomials $\phi(B) = 1 - \sum_{i=1}^{p} \phi_i B^i$ and $\theta(B) = 1 - \sum_{i=1}^{q} \theta_i B^i$, we can obtain, by long division, that

$$\frac{\theta(B)}{\phi(B)} = 1 + \psi_1 B + \psi_2 B^2 + \ldots =: \psi(B)$$

and

$$\frac{\phi(B)}{\theta(B)} = 1 - \pi_1 B - \pi_2 B^2 - \ldots =: \pi(B).$$

For instance, if $\phi(B) = 1 - \phi_1 B$ and $\theta(B) = 1 - \theta_1 B$, then

$$\psi(B) = \frac{1 - \theta_1 B}{1 - \phi_1 B} = 1 + (\phi_1 - \theta_1) B + \phi_1(\phi_1 - \theta_1) B^2 + \phi_1^2 (\phi_1 - \theta_1) B^3 + \ldots,$$

and

$$\pi(B) = \frac{1 - \phi_1 B}{1 - \theta_1 B} = 1 - (\phi_1 - \theta_1) B - \theta_1(\phi_1 - \theta_1) B^2 - \theta_1^2 (\phi_1 - \theta_1) B^3 + \ldots.$$

It follows directly from the definitions that $\psi(B)\pi(B) = 1$. Recall the representation of an $ARMA(p, q)$ model as $\phi(B) r_t = \phi_0 + \theta(B) a_t$. The MA representation of an $ARMA(p, q)$ model is given by

$$r_t = \phi(B)^{-1}\phi_0 + \phi(B)^{-1}\theta(B) a_t = c_0 + \psi(B) a_t = c_0 + a_t + \sum_{i=1}^{\infty} \psi_i a_{t-i}, \tag{2}$$

where we have set $c_0 := \phi(B)^{-1}\phi_0$, and we can easily see that $\phi(B)^{-1}\phi_0 = \frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ using the following argument: $\phi(B)^{-1}\phi_0 = \frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ if and only if $\phi_0 = \phi(B)\frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ where we have applied the operator $\phi(B)$ on the left and right hand side of the previous equality. Observe that, for a constant c, Bc is a constant, which further implies that $B^n c = c$ for any integer n. Setting $c := \frac{\phi_0}{1-\phi_1-\cdots-\phi_p}$ we would obtain that $\phi(B)c = c - \phi_1 c - \phi_2 c - \ldots - \phi_p c = c(1 - \phi_1 - \ldots - \phi_p) = \phi_0$, hence proving the result.

Remark 2.1. *In writing the representation* (2)*, we are using the fact that ϕ has no complex root within the unit circle. If ϕ had complex roots within or at the boundary of the unit circle, then the series r_t would take the form*

$$r_t = \phi(B)^{-1}\phi_0 + \phi(B)^{-1}\theta(B) a_t = c_0 + \psi(B) a_t = c_0 + \sum_{i=-\infty}^{\infty} \psi_i a_{t-i}, , \tag{3}$$

where $\psi_i \neq 0$ for some $i < 0$, so that the return at time t would also depend on future innovations a_s for $s > t$.

The representation (2) shows explicitly the impact of the past shock $a_{t-i} (i > 0)$ on the current return r_t. The coefficients ψ_i's are referred to as the *impulse response function* of the ARMA model. For a weakly stationary series, the ψ_i coefficients decay exponentially as i increases, consistently with intuition: the effect of shock a_{t-i} on the return r_t should decrease over time. Thus, the shock a_{t-i} does not have a permanent impact on the series.

Remark 2.2. *Suppose we want to recover the innovations as past observations of r_t. Then we look for a AR representation of the $ARMA(p,q)$ model, i.e., starting from*

$$\phi(B)r_t = \phi_0 + \theta(B)a_t, \tag{4}$$

where we recall that

$$\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i, \qquad \theta(z) := 1 - \sum_{j=1}^{q} \theta_j z^j.$$

If we multiply LHS and RHS of (4) by $\theta(B)^{-1}$, we obtain

$$a_t = \theta(B)^{-1}\phi(B)r_t - \theta(B)^{-1}\phi_0.$$

Then, we can write

$$a_t = d_0 + \pi(B)r_t = d_0 + r_t - \sum_{i=1}^{\infty} \pi_i r_{t-i}.$$

where $d_0 = -\theta(B)^{-1}\phi_0$. The above representation holds only if the ARMA model is invertible, i.e., $\theta(B)$ does not have any complex root within or on the unit circle. If $\theta(B)$ has complex roots within or on the unit circle, then the representation would take the form $a_t = d_0 + \pi(B)r_t = d_0 - \sum_{i=-\infty}^{\infty} \pi_i r_{t-i}$, where $\pi_i \neq 0$ for some $i < 0$. Hence, the shock (innovation) at time t would depend on future returns r_s, $s > t$.